

Assignment-21.3

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Batch 41

Task-1:AI-Assisted Pair Programming

PROMPT:

Give code for expense tracker

CODE:

```
Lab21.py > ...
1 # basic_expense_tracker.py
2 expenses = []
3 while True:
4     print("\n1. Add expense\n2. View all\n3. Exit")
5     choice = input("Select: ")
6     if choice == "1":
7         amount = input("Amount: ")
8         category = input("Category: ")
9         date = input("Date (YYYY-MM-DD): ")
10        expenses.append({
11            "amount": amount,
12            "category": category,
13            "date": date
14        })
15
16        print("Added!")
17    elif choice == "2":
18        for e in expenses:
19            print(e)
20    elif choice == "3":
21        break
22    else:
23        print("Invalid choice!")
```

OUTPUT:

```
.\venv\Scripts\Activate.ps1
(.venv) PS C:\Users\Pranitha Rao\OneDrive\Desktop\AI ASSISTED CODING> & "C:/Users/Pranitha Rao/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Pranitha Rao/OneDrive/Desktop/AI ASSISTED CODING/Lab21.py"

1. Add expense
2. View all
3. Exit
Select: 1
Amount: 12000
Category: Travel
Date (YYYY-MM-DD): 2026-04-04
Added!

1. Add expense
2. View all
3. Exit
Select: 2
{'amount': '12000', 'category': 'Travel', 'date': '2026-04-04'}

1. Add expense
2. View all
3. Exit
Select: 
```

PROMPT:

Write a Python program for a simple Expense Tracker Application that allows a user to record daily expenses along with categories (like food, travel, etc.). First, generate a basic version of the program with simple input and storage. Then review the code and identify its limitations, such as lack of input validation or structure. After that, improve the program by adding features like monthly expense summary, proper input validation, and organizing the code into functions for better readability. Finally, provide both the initial and improved versions and explain how the changes make the program more efficient and user-friendly.

CODE:

```
25 # Improved Expense Tracker
26 expenses=[]
27 def add_expense():
28     try:
29         amount=float(input("Enter amount: "))
30     except ValueError:
31         print("Invalid amount!")
32         return
33     category=input("Enter category: ")
34     date=input("Enter date (YYYY-MM-DD): ")
35     expenses.append({"amount":amount,"category":category,"date":date})
36     print("Expense added successfully!")
37 def view_expenses():
38     if not expenses:
39         print("No expenses found.")
40         return
41     print("\nExpense List:")
42     for e in expenses:
43         print(f"{e['date']} | {e['category']} | ₹{e['amount']}")
44 def monthly_summary():
45     month=input("Enter month (YYYY-MM): ")
46     total=0
47     for e in expenses:
48         if e["date"].startswith(month):
49             total+=e["amount"]
50     print(f"Total expense for {month}: ₹{total}")
51 def main():
52     while True:
53         print("\n--- Expense Tracker ---")
54         print("1. Add Expense")
55         print("2. View Expenses")
56         print("3. Monthly Summary")
57         print("4. Exit")
```

```
58         choice=input("Enter choice: ")
59         if choice=="1":
60             add_expense()
61         elif choice=="2":
62             view_expenses()
63         elif choice=="3":
64             monthly_summary()
65         elif choice=="4":
66             break
67         else:
68             print("Invalid choice!")
69     main()
```

OUTPUT:

```
(.venv) PS C:\Users\Pranitha Rao\OneDrive\Desktop\AI ASSISTED CODING> & "C:/Users/Pranitha Rao/AppData/Local/Microsoft/WindowsApps/python3.11.exe" "C:/Users/Pranitha Rao/OneDrive/Desktop/AI ASSISTED CODING/Lab21.py"
Enter choice: 1
Enter amount: 12345
Enter category: Food
Enter date (YYYY-MM-DD): 2026-04-06
Expense added successfully!

--- Expense Tracker ---
1. Add Expense
2. View Expenses
3. Monthly Summary
4. Exit
Enter choice: 1
Enter amount: 1234
Enter category: Travel
Enter date (YYYY-MM-DD): 2026-04-06
Expense added successfully!

--- Expense Tracker ---
1. Add Expense
2. View Expenses
3. Monthly Summary
4. Exit
Enter choice: 2

Expense List:
2026-04-06 | Food | ₹12345.0
2026-04-06 | Travel | ₹1234.0

--- Expense Tracker ---
1. Add Expense
2. View Expenses
3. Monthly Summary
4. Exit
Enter choice: 3
Enter month (YYYY-MM): 2026-04
Total expense for 2026-04: ₹13579.0
```

JUSTIFICATION:

The basic Expense Tracker allows users to add and view expenses but lacks validation, structure, and analysis features. The improved version uses functions, input validation, and a monthly summary, making the program more organized, reliable, and user-friendly.

Task 2 – Version Control Integration

```
pranitha@pranitha-750XGK:~/Downloads/assistantcoding-2020631770646292-3-0015$ cd /home/pranitha/Downloads/grade-calculator
Switched to branch 'main'.

pranitha@pranitha-750XGK:~/Downloads/assistantcoding$ cd /home/pranitha/Downloads/pranitha && git checkout -b feature-ai-grade && cd /home/pranitha/Downloads/grade-calculator &
pranitha@pranitha-750XGK:~/Downloads/assistantcoding$ cd /home/pranitha/Downloads/pranitha && git checkout -b feature-ai-grade && cd /home/pranitha/Downloads/grade-calculator &
pranitha@pranitha-750XGK:~/Downloads/assistantcoding$ /cd /home/pranitha/Downloads/pranitha && git checkout -b feature-ai-grade && cd /home/pranitha/Downloads/grade-calculator &
pranitha@pranitha-750XGK:~/Downloads/grade-calculator$
Switched to a new branch 'feature-ai-grade'

pranitha@pranitha-750XGK:~/Downloads/assistantcoding$ cd /home/pranitha/Downloads/grade-calculator &
marks = List(map(int, input("Enter marks: ").split()))
total = sum(marks)
avg = total / len(marks)
grade = ""
if avg >= 90:
    grade = 'A'
elif avg >= 80:
    grade = 'B'
elif avg >= 70:
    grade = 'C'
elif avg >= 60:
    grade = 'D'
elif avg >= 50:
    grade = 'E'
else:
    grade = 'F'
print("Average:", avg)
print("Percentage", percentage)
print("Grade", grade)

pranitha@pranitha-750XGK:~/Downloads/grade-calculator$
```

PROMPT:

Add grade classification and percentage calculation to the program.

CODE:

```
student_grade_calculator.py > ...
1 marks = list(map(int, input("Enter marks: ").split()))
2 total = sum(marks)
3 avg = total / len(marks)
4 percentage = avg # since marks are out of 100
5 if avg >= 90:
6     grade = 'A'
7 elif avg >= 80:
8     grade = 'B'
9 elif avg >= 70:
10    grade = 'C'
11 elif avg >= 60:
12    grade = 'D'
13 elif avg >= 50:
14    grade = 'E'
15 else:
16    grade = 'F'
17 print("Average:", avg)
18 print("Percentage:", percentage)
19 print("Grade:", grade)
```

OUTPUT:

```
pranitha@pranitha-750XGK:~/Downloads/quiz-managements git checkout -b feature-score
Switched to a new branch 'feature-score'

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git add .
Switched to a new branch 'feature-score'

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git commit -m "Added score calculation"
[feature-score 8ab91cd] Added score calculation
1 file changed, 12 insertions(+), 1 deletion(-)

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git checkout main
Switched to branch 'main'
Swtchjnt to branch 'main'

pranitha@pranitha-750XGK:~/L1pmggr/cwentscad main
Quettted to branch 'main'
accepttd HE00 -> main, feature-score

├─ 2+2 4
│   Correct
└─ Final Score: 1

pranitha@pranitha-750XGK:~/L1pmggr/cwentscad git log --oneline --decorate --graph --all --max-count=5
• 8ab91cd (HEAD -> 0 main, feature-score)
└─ > Added score calculation *****
+ 2f4a7d1 Initial quiz management system
```

JUSTIFICATION:

The project uses Git for version control to manage changes efficiently. A separate feature branch was created to add score calculation functionality without affecting the main code. After implementing and testing the feature, it was committed with a meaningful message and merged into the main branch. This approach ensures better code management, safe development, and easy tracking of changes, making the project more organized and professional.

Task 3 – AI for Commit Message Generation

```
import re

password = input("Enter password: ")

if len(password) < 8:
    print("Weak password")
elif not re.search(r'[!@#$$%^&*]', password):
    print("Add special character")
else:
    print("Strong password")
```

PROMPT:

Generate a commit message for adding password validation with length and special character rules.

AI Committed Message:

```
750XGK:~/Downloads/aiassistantcoding-20260312T084629Z-3-001$
av-750XGK:~/Downloads/aiassistantcoding-20260312T084629Z-3-001$ git commit -m "feat: add password validation with minimum length and special charac
```

Human Version:

Added strong password validation rules.

JUSTIFICATION:

AI more structured (uses “feat:” convention), Human → less descriptive.

Task 4 – Pair Programming Simulation

```
questions = {
    "Capital of India?": "Delhi",
    "2+2?": "4"
}

for q in questions:
    ans = input(q + " ")
    if ans == questions[q]:
        print("Correct")
    else:
        print("Wrong")
```

AI Prompt (Student B):

Add score calculation and input validation to the quiz program.

Final Code:

```
questions = {
    "Capital of India?": "Delhi",
    "2+2?": "4"
}
score = 0
for q in questions:
    ans = input(q + " ").strip()
    # Input validation (check empty input)
    if ans == "":
        print("Invalid input! Please enter an answer.")
        continue
    if ans.lower() == questions[q].lower():
        print("Correct")
        score += 1
    else:
        print("Wrong")
print("Final Score:", score)
```

Git WorkFlow:

```
pranitha@pranitha-750XGK:~/Downloads/quiz-managements git checkout -b feature-score
Switched to a new branch "feature-score"

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git add .
Switched to a new branch "feature-score"

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git commit -m "Added score calculation"
[feature-score 8ab91cd] Added score calculation
> file changed, 12 insertions(+), 1 deletion(-)

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git checkout main
Switched to branch 'main'
Updating 2f47ad1..8ab91cd
Fast-forward
 quiz.py | 12 +: 11 insertions(+), 1 deletion(-)
-Wrong
Final Score: 1
Wrong
Final Score: 1

pranitha@pranitha-750XGK:~/Downloads/quiz-managements git log --oneline --decorate --graph --all --max-count=5
* 8ab91cd (HEAD <> main, feature-score) Added score calcul
└─> 2f47ad1 Initial quiz management system

• pranitha@pranitha-750XGK:~/Downloads/quiz-managements
```

OUTPUT:

```
pranitha@pranitha-750XGK:~/Downloads$ /usr/bin/python3 /home/pranitha/Downloads/assistantcoding
2020631270846292-3-001/assistantcoding65.py
Capital of India? Y
Wrong
2+2? 4
Correct
Final Score: 1
pranitha@pranitha-750XGK:~/Downloads$
```

JUSTIFICATION:

The program was enhanced by adding score calculation and input validation, making it more accurate and user-friendly. Using Git, changes were developed in a separate branch and then merged into the main branch, ensuring better code management and safe updates.